

SMART LIGHTING

Army Institute of Technology

Department of Information Technology

Under the guidance of Prof. Snehal Shinde Dhomade

by

4115 Darshan

4120 Harsh Pundir

4121 Harshwardhan

4122 Himanshu

INTRODUCTION

Smart lighting is a key component of modern home automation.

- Home automation demand increasing since networking era
- Existing smart lighting systems are costly and complex
- Need for affordable and easy-to-install smart switch
- Smartphone control becoming standard feature

Limitations of conventional lighting systems

- Cannot control lights remotely
- High-voltage lights difficult to interface with controllers
- No systems that learn user lighting patterns
- Energy wasted when lights left ON in empty rooms

PROBLEM STATEMENT

The Problem Current lighting systems are inefficient. Conventional setups require manual effort, while existing smart solutions are often expensive, complicated, and lack the ability to learn user patterns. This leads to lights being left on unnecessarily, wasting energy.

The Solution There is a specific need for an **affordable, easy-to-install smart switch**. This device should offer remote control capabilities and intelligently automate lighting by detecting room occupancy and learning user habits.

OBJECTIVES OF PROJECT

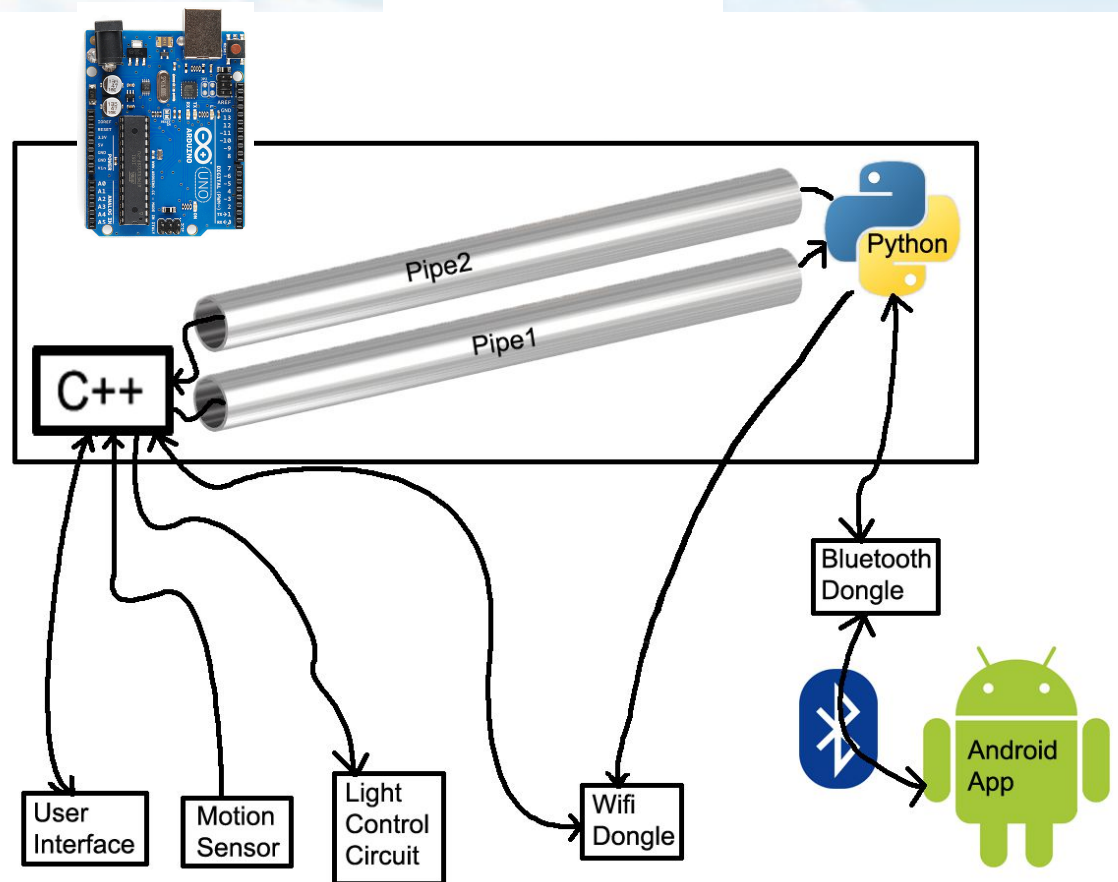
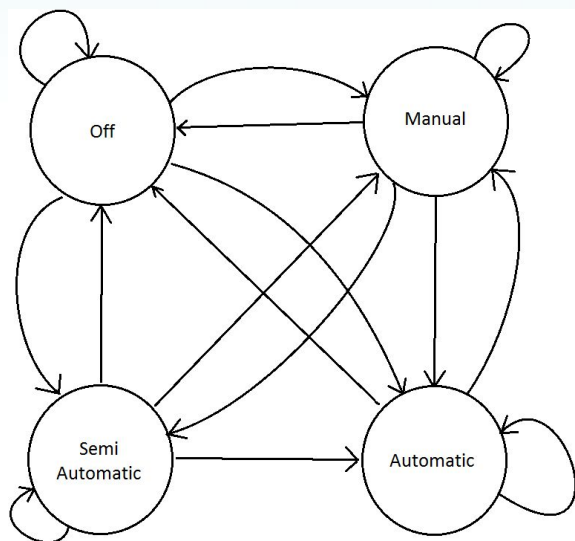
Goal: Develop a smart lighting switch prototype

- Replace conventional wall light switch
- Enable smartphone & manual control
- Detect motion and save energy
- Learn and predict lighting usage patterns

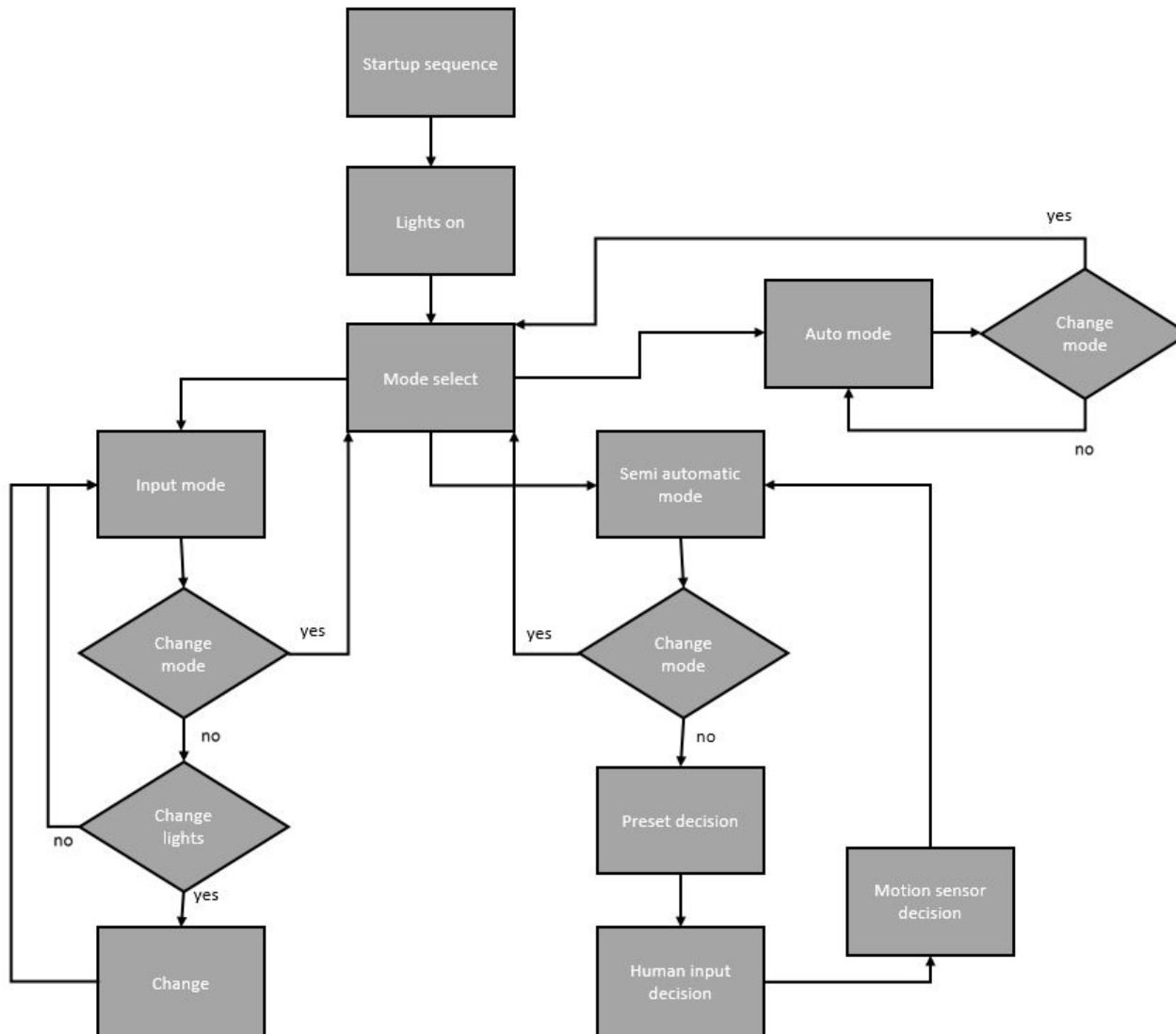
LITERATURE SURVEY

Sr.No	Paper Name/title	Author Name/Year	Methodology	Future Scope/short comings	Results
1	Smart Lighting System Using Arduino	Authors: S. Jung, S. Lee, et al. Year: 2022	Hardware: Arduino Uno controller with Ultrasonic & IR sensors for presence detection. Logic: Uses a "Person-Counter" mechanism to track occupancy and toggle lights accordingly.	Shortcomings: Relies heavily on line-of-sight sensors; if the sensor is obstructed, the system fails to detect presence. Future Scope: Integration with dimming control based on sunlight intensity (LDR).	Energy Savings: Achieved a 50% reduction in energy consumption (0.4 kWh to 0.2 kWh daily) by eliminating manual operation errors.
2	Bluetooth Based Home Automation System	Authors: Satish Kumar Singh et al. Year: 2019	Comms: Uses HC-05 Bluetooth module interfaced with Arduino to control home appliances via a custom Android App. Control: Simple ON/OFF switching via smartphone GUI.	Shortcomings: Range Limitation: Bluetooth restricts control to within ~10 meters (room-level only). Security: Basic Bluetooth protocols are susceptible to unauthorized pairing if not secured.	Cost-Efficiency: Proven to be the most cost-effective wireless solution compared to ZigBee or Wi-Fi for single-room automation, validating the low-budget approach.
3	IoT-Based Smart Lighting System (ISLS)	Authors: Liangliang Ma Year: 2024	Connectivity: Uses NodeMCU (Wi-Fi) to connect lighting systems to the Cloud (IoT). Algorithm: "Logic-based" method that uses decision trees to determine light levels based on time of day and occupancy.	Shortcomings: Complexity: Requires constant internet connection and cloud server setup, increasing initial cost and complexity. Latency: Dependent on network speed; internet outages disable remote control.	User Comfort: Demonstrated that automated "Adaptive Lighting" (changing brightness based on needs) improves user visual comfort and reduces eye strain significantly.

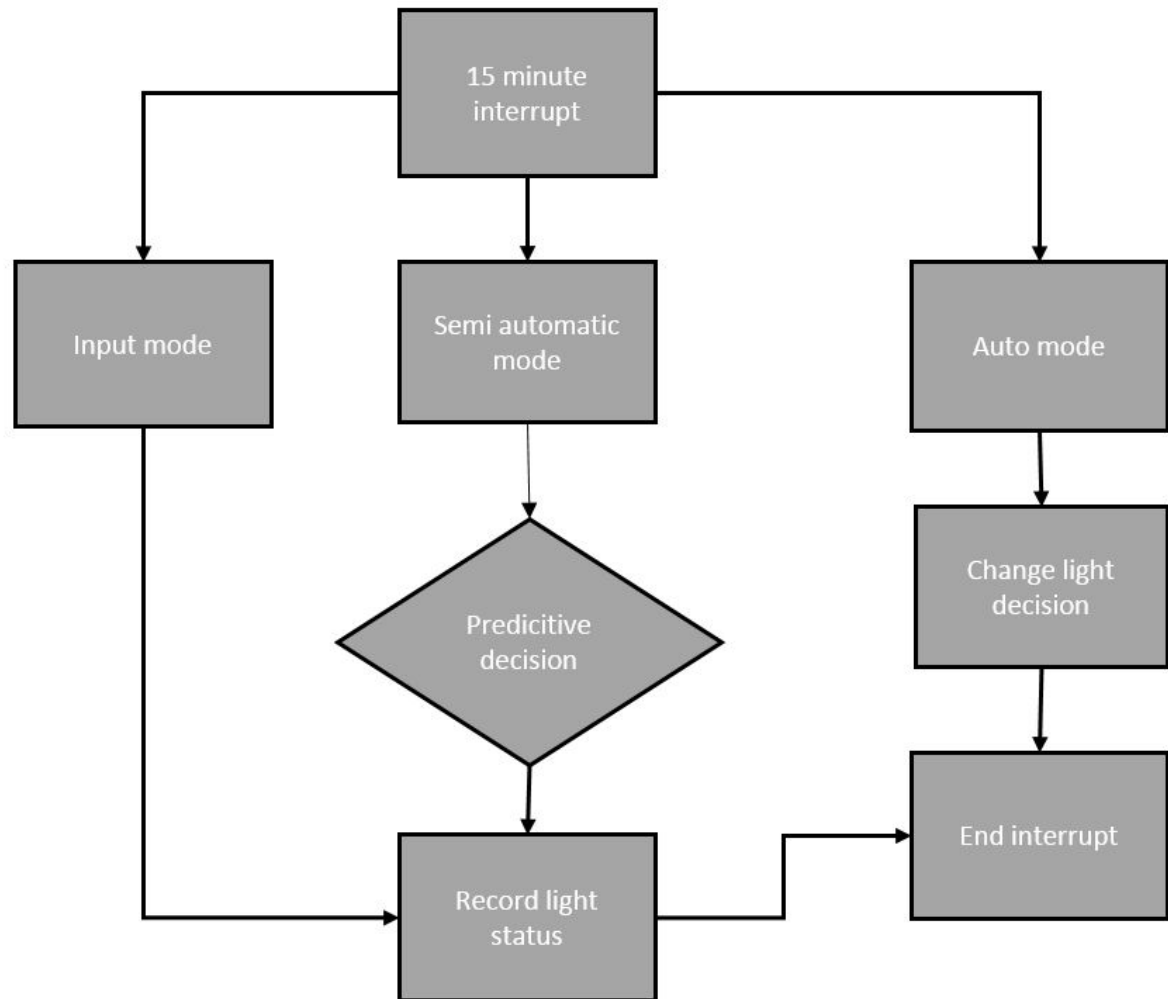
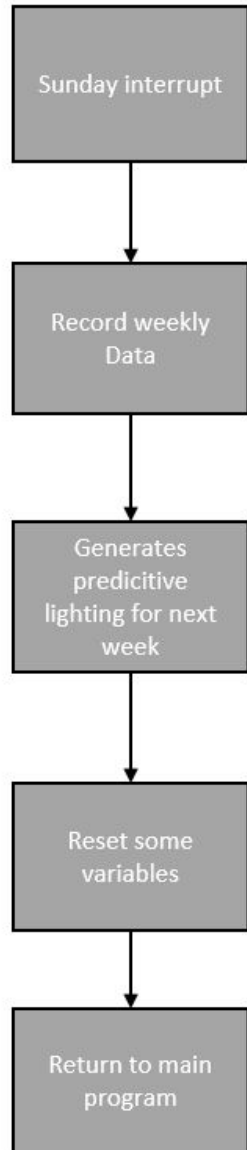
METHODOLOGY



FLOWCHART



FLOWCHART



LIMITATIONS

Limitations of the Proposed Smart Lighting System

- Prototype not integrated into a single compact wall-switch unit
- Prediction algorithm tested only in simulation (MATLAB), not real deployment
- Internet control removed due to development constraints
- Electrical safety certification not implemented in prototype
- Limited prediction accuracy (~60% for weekly events)
- Designed for single-room lighting control only

DISCUSSION

◆ Interpretation of Results

- The smart lighting system successfully demonstrates automation using motion detection and user input
- System shows basic learning capability for lighting patterns (via prediction logic)
- Energy saving is achieved by reducing unnecessary ON time of lights

◆ Challenges Faced

- Integration of all components into a compact hardware unit
- Limited real-world testing (prediction tested only in simulation)
- Sensor accuracy issues (motion detection reliability)
- Removal of internet control due to development constraints
- Achieving higher prediction accuracy (~60% currently)

◆ Key Observations

- Automation significantly improves user convenience
- Even basic prediction improves efficiency over manual systems
- System is cost-effective compared to existing smart solutions
- Works well for single-room implementation, but needs scaling

APPLICATION & FUTURE SCOPE

Applications of Smart Lighting System

- Smart home automation lighting control
- Energy-efficient lighting in homes and offices
- Security lighting (presence simulation when away)
- Automated room occupancy lighting
- Assistive lighting for elderly or disabled users
- Smart alarm system using light patterns

Future Scope and Enhancements

- Integration into standard wall switch form factor
- Cloud and mobile app remote control via Internet
- AI-based learning for higher prediction accuracy
- Multi-room and whole-home lighting networks
- Voice assistant integration (Alexa / Google Home)
- Energy monitoring and usage analytics

CONCLUSION

Smart switch replacing traditional switch

- Controlled via buttons or Android app
- Uses Arduino controller
- Motion sensor detects room occupancy
- Relay circuit controls lights

EXPECTED RESULTS

◆ Expected Outcomes

- Development of a fully functional smart lighting switch prototype
- Seamless control via manual switch + smartphone interface
- Reliable motion-based automatic lighting system

◆ Performance Improvements

- Reduction in **energy consumption** due to automation
- Faster response time in switching compared to manual systems
- Improved user comfort and convenience

◆ Accuracy & Efficiency

- Prediction accuracy expected to improve with **real-time data learning**
- Efficient lighting usage based on **occupancy and user habits**
- System aims for higher accuracy (>80%) in future versions

REFERENCES

- **R. Heukels, “Predicting User Behavior Using Transition Probability,” University of Twente, 2015.**
 - • Provides the mathematical model for predicting user behavior from state transitions.
 - • Used as the basis for the lighting pattern recognition and prediction algorithm in this project.
- **Smart Lighting and Home Automation Literature**
 - • Industry and research sources describing smart lighting architectures and control methods.
 - • Provided background on motion sensing, remote control, and energy-efficient lighting automation concepts.